CLASSIFICATION & LOGISTIC REGRESSION

Training set $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, but now

$y^{(i)} \in \{1, 2, \dots, k\}$

$y^{(i)} \in \{\text{'dog'}, \text{'cat'}, \dots\}$

$\in \{\text{'spam'}, \text{'not-spam'}\}$

LABELS are discrete in this case.

If you have a regression, it is possible to transform it into a classification problem:

For example, $y \Rightarrow$ grade in a class out of 10 point.

$y \in \mathbb{R} \in [0, 10]$

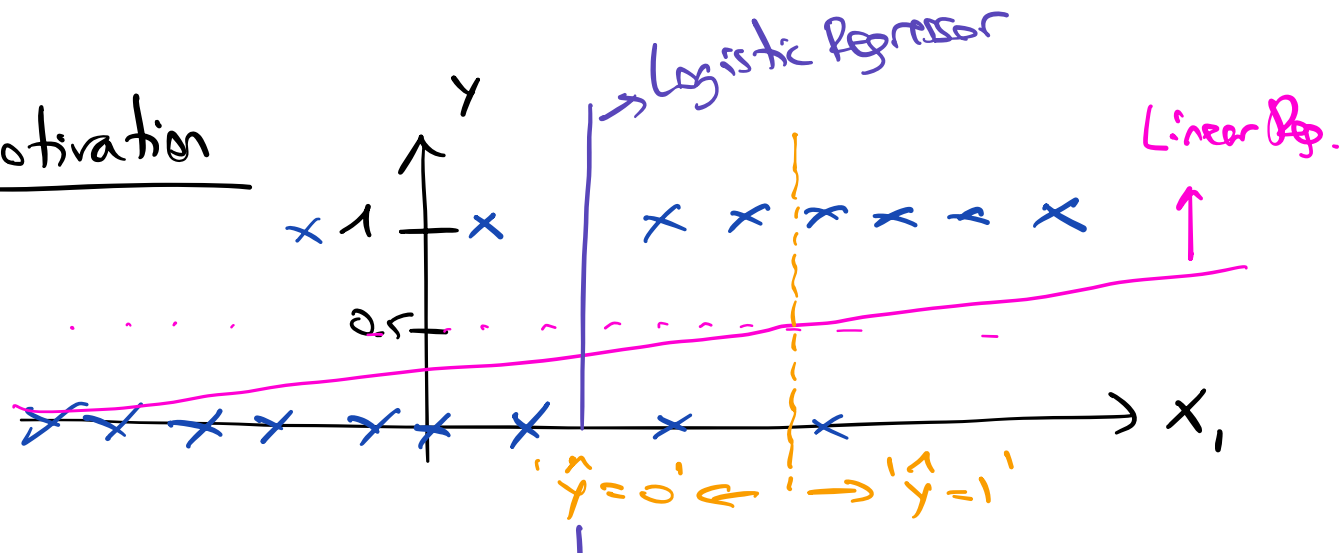
$y \in \{A^+, A, B^+, B, C, D\}$

simplest (but very common case) in which $y \in \{0, 1\}$.

<u>BINARY CLASSIFICATION</u> problem.

Normally, by convention $\begin{cases} y = 0 \rightarrow \text{'negative' class} \\ y = 1 \rightarrow \text{'positive' class} \end{cases}$

Motivation



Logistic Regression model

→ Linear model
→ For binary classification
→ The simplest class. model.

Prediction:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} = \sigma(\theta^T x) \in (0,1)$$

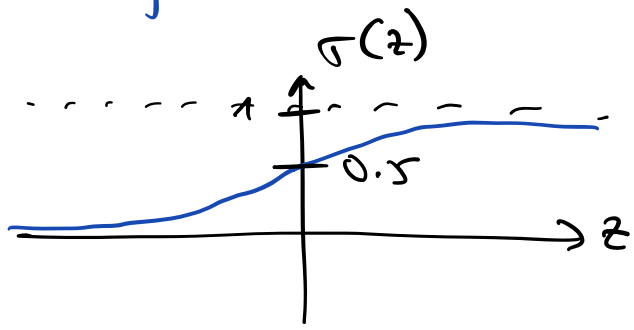
Sigmoid or logistic function

Linear Reg ⇒ $h_\theta(x) = \theta^T x$

$$\theta^T x = \theta_0^{(x_0=1)} + \sum_{j=1}^{n} \theta_j x_j$$

Sigmoid function $\sigma(z)$:

$$\sigma(z) = \frac{1}{1+e^{-z}} \Rightarrow$$



$\sigma(0) = 0.5$

$\lim_{z \to \infty} \sigma(z) = 1$ $\quad$ $\lim_{z \to -\infty} \sigma(z) = 0$

$\sigma(z)$ → Nice for opt. algorithms (it's convex)
→ Special case of a broader family of algorithms called 'Generalized Linear Models (GLM)'

Interpretation of $h_\theta(x)$:

- Probabilistic assumptions

$$\begin{cases} p(y=1 \mid x;\theta) = h_\theta(x) \\ p(y=0 \mid x;\theta) = 1 - h_\theta(x) \end{cases}$$

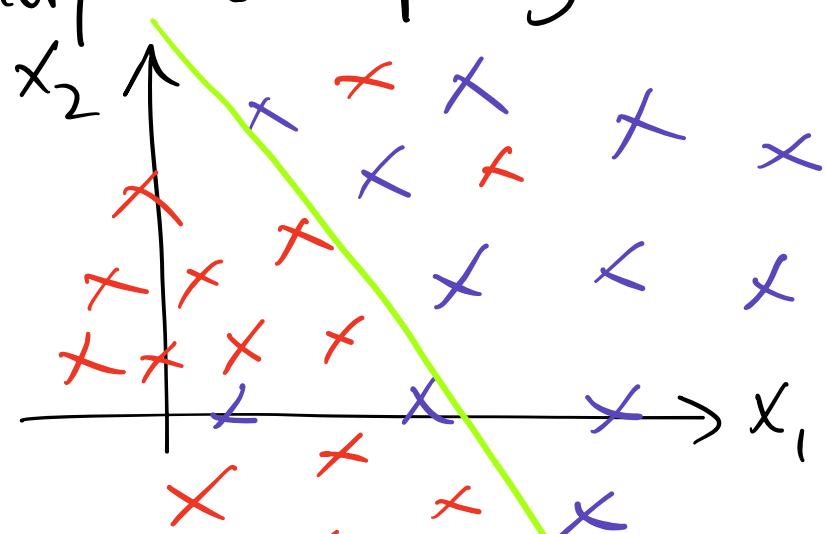
To make a prediction:

$$\hat{y} = \begin{cases} 1, & \text{if } h_\theta(x) \geq 0.5 \\ 0, & \text{oth.} \end{cases}$$

Log. Regression is a <u>SOFT</u> classifier, because it outputs probabilities.

There are other classifiers that output directly the class (or the label), and these are called <u>HARD</u> classifiers (e.g., SVM).

You can show that the decision boundary that you get is linear. (HW1)



→ Log. Reg.
All xs s.t.:
→ $h_\theta(x) = 0.5$

How do we find the best value of θ?

$$p(y \mid x;\theta) = (h_\theta(x))^{y} \cdot (1 - h_\theta(x))^{1-y}$$

if $y=1 \rightarrow p(y=1 \mid x;\theta) = h_\theta(x)$

if $y=0 \rightarrow p(y=0 \mid x;\theta) = 1 - h_\theta(x)$

→ This is for one training example.

Taking all training examples and <u>assuming independance between them</u>:

↳ This assumption is always made in ML.

$$p(\bar{y} \mid X;\theta) \overset{\text{IND}}{=} \prod_{i=1}^{m} p(y^{(i)} \mid x^{(i)};\theta)$$

$$\begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix} \qquad \begin{bmatrix} - & x^{(1)T} & - \\ & \vdots & \\ - & x^{(m)T} & - \end{bmatrix}$$

$$= \prod_{i=1}^{m} (h_\theta(x^{(i)}))^{y^{(i)}} (1 - h_\theta(x^{(i)}))^{1-y^{(i)}}$$

$= L(\theta) \Rightarrow$ Likelihood Function

We are going to look for the θ that maximizes $L(\theta) \Rightarrow$ Max. Likelihood (ML)

$$\boxed{\hat{\theta} = \arg\max_{\theta} L(\theta)}$$

$$L(\theta) = \prod_{i=1}^{m} h_\theta(x)^{y} \left(1 - h_\theta(x)\right)^{1-y}$$

$$\left.\begin{array}{l} y^{(1)} = 0 \\ y^{(2)} = 1 \\ y^{(3)} = 1 \end{array}\right\} \quad L(\theta) = \left(1 - h_\theta(x^{(1)})\right) \cdot h_\theta(x^{(2)}) \cdot h_\theta(x^{(3)})$$

To max. $L(\theta)$, ideally I should have:

$$\left\{\begin{array}{l} h_\theta(x^{(1)}) = 0 = \dfrac{1}{1 + e^{-\theta^T x^{(1)}}} \\ h_\theta(x^{(2)}) = 1 \rightarrow \dfrac{1}{\quad} \end{array}\right.$$

$$h_\theta(x^{(3)}) = 1 \qquad \frac{1}{1+e^{-\theta^T x^{(2)}}}$$

$$\frac{1}{1+e^{-\theta^T x^{(3)}}}$$

$$\hat{\theta} = \arg\max_\theta L(\theta)$$

$$= \arg\max_\theta \underbrace{\log(L(\theta))}_{\ell(\theta)}$$

(log-likelihood function)

$$\ell(\theta) = \log\left(\prod(\quad)(\quad)\right)$$

$$= \sum_{i=1}^{m}\Big(y^{(i)}\log h_\theta(y^{(i)}) +$$

$$+ (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\Big)$$

$$\hat{\theta} = \arg\max_\theta \ell(\theta)$$

↳ How do I find θ?

$$= \arg\min_{\theta} \boxed{-\ell(\theta)}$$

→ I can use the GD algorithm to solve for θ.

Until convergence

For $j = 0$ to n

$$\theta_j := \theta_j - \alpha \underbrace{\frac{\partial}{\partial \theta_j}\left(-\ell(\theta)\right)}$$

The derivation in the LAB session.

<u>Update rule</u>

$$\theta_j := \theta_j - \alpha \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

→ Identical expression to the one found in Linear Regression.

Difference since h_θ

But there is a key difference, since in this case $h_\theta(x) = \sigma(\theta^T x)$.

$$-\ell(\theta) = -\sum\Big(\Big(y^{(i)} \log \overbrace{h(x^{(i)})}^{p(y=1|x;\theta)} + (1-y^{(i)}) \log \underbrace{(1-h(x^{(i)}))}_{p(y=0|x;\theta)}\Big)$$

→ Binary cross-entropy loss!!

(This is the loss use in Deep Learning for binary class. problems).

Multi-Class classification

- In this case, $y \in \{1, 2, \ldots, K\}$

  Example: e-mails, they could be classified as "spam, work-related, other"

- One option can be to use a

binary classifier, and take a one-vs-all approach.

$y \in \{1, 2, 3\}$

1) $Y_1 = \begin{cases} 1, & \text{if } y = 1 \\ 0, & \text{if } y \in \{2,3\} \end{cases} \rightarrow$ Fit a model

$Y_2 = \begin{cases} 1, & \text{if } y = 2 \\ 0, & \text{oth.} \end{cases} \rightarrow$ Fit a model

$Y_3 = \begin{cases} 1, & \text{if } y = 3 \\ 0, & \text{oth} \end{cases} \rightarrow$ Fit a model

<u>Prediction</u>: The one with the highest probability.

- In gel, use a model that can handle multi-class setting.

- <u>SOFTMAX REGRESSOR</u>

  - An extension of log. regression

- Model the problem as a <u>multinomial distribution</u>

- $$\left.\begin{aligned} p(y=1 \mid x;\theta) &= \phi_1 \\ p(y=2 \mid x;\theta) &= \phi_2 \\ &\vdots \\ p(y=k \mid x;\theta) &= \phi_k \end{aligned}\right\} \quad \sum_{j=1}^{k} \phi_j = 1$$

- <u>Parameters</u>: $\theta_1, \theta_2, \ldots, \theta_k \in \mathbb{R}^{n+1}$

$$\phi_i = f(\theta_i^T x)$$

I cannot choose $\phi_i = \theta_i^T x \notin (0,1)$

We will use the <u>softmax function</u>

$$\text{softmax}(t_1, t_2, \ldots, t_k) = \begin{bmatrix} \dfrac{e^{t_1}}{\sum_{i=1}^{k} e^{t_i}} \\ \vdots \\ \dfrac{e^{t_k}}{\sum_{i=1}^{k} e^{t_i}} \end{bmatrix} \begin{matrix} \in (0,1) \\ \\ \in (0,1) \end{matrix}$$

$$\sum(\quad) = 1$$

$$\begin{bmatrix} p(y=1|x;\theta_1) \\ \vdots \\ p(y=k|x;\theta_k) \end{bmatrix} = \begin{bmatrix} \phi_1 \\ \phi_2 \\ | \\ \phi_k \end{bmatrix} = \text{softmax}(\theta_1^T x, \ldots, \theta_k^T x)$$

$$\phi_i = p(y=i|x;\theta_i) = \frac{e^{\theta_i^T x}}{\sum_{j=1}^{k} e^{\theta_j^T x}}$$

Assuming you know $\theta_1, \theta_2, \ldots, \theta_k$:

Prediction will be $[\phi_1, \phi_2, \ldots, \phi_k]$

You get a set of k probabilities, & the prediction will be the class with the highest prob.

$$[\phi_1 = 0.1, \phi_2 = 0.8, \phi_3 = 0.1] \Rightarrow \underline{\hat{y} = 2}$$

If $y \in \{0,1\}$ $\Rightarrow \begin{bmatrix} \dfrac{e^{\theta_1^T x}}{e^{\theta_1^T x} + e^{\theta_2^T x}} \end{bmatrix}$

$$\boxed{\frac{e^{\theta_2^T x}}{e^{\theta_1^T x} + e^{\theta_2^T x}}}$$

- How do we find the best θs?

<u>Likelihood function</u>

$$p(y|x;\theta) \underset{\text{IND}}{=} \prod_{i=1}^{m} p(y^{(i)}|x^{(i)};\theta)$$

$$= \prod_{i=1}^{m} \overbrace{\frac{e^{\theta_{y^{(i)}}^T x^{(i)}}}{\sum_{j=1}^{k} e^{\theta_j^T x^{(i)}}}}^{\phi_{y^{(i)}}}$$

$$= L(\theta)$$

$$\hat{\theta} = \arg\min_{\theta} -L(\theta) = \arg\min_{\theta} \underbrace{-\ell(\theta)}_{\text{log-likelihood}}$$

$\rightarrow$ I can again use GD.

→ if you work it out, you will arrive to minimizing what's call the <u>cross-entropy loss</u> (ℓ_{ce})

$$\ell_{ce} : \mathbb{R}^k \times \{1, 2, \ldots, k\} \to \mathbb{R}_{\geq 0}$$

$$\ell_{ce}((t_1, t_2, \ldots, t_k), y) = -\log\left(\frac{e^{t_y}}{\sum_{j=1}^{k} e^{t_j}}\right)$$

$$-\ell(\theta) = \sum_{i=1}^{m} \ell_{ce}\left(\left(\theta_1^T x^{(i)}, \ldots, \theta_k^T x^{(i)}\right), y^{(i)}\right)$$